EXAM PAPERS PRACTICE

OCR AS and A Level Economics (H060, H460) Topic Questions

Question Paper

Main Topic	**Microeconomics (Component 01)**
Sub-Topic	**Market Structure**

To be used by all students and teachers for OCR AS and A Level Economics (H060, H460), for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ______________________

Date : ______________________

Time : ______________________

Total Marks Available : _____

Total Marks Received : _____

1 * Firms usually take into account the income and cross elasticities of demand for their products when setting their prices.

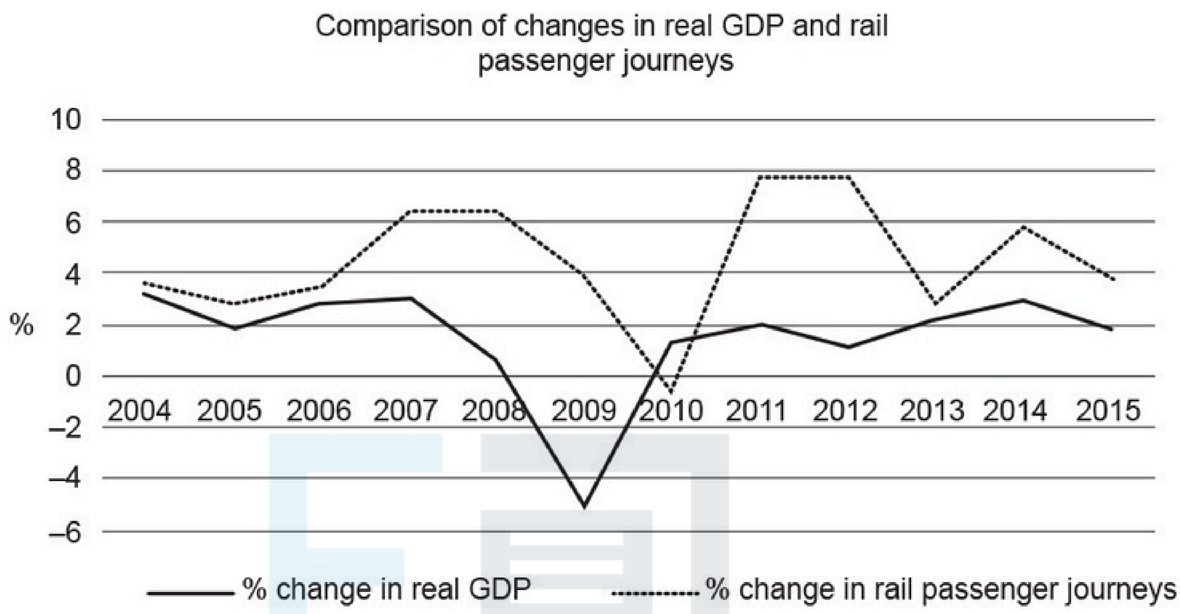
Evaluate whether a firm which produces a product that has positive income elasticity of demand and positive cross elasticity of demand should lower the price of the product.

[25]

2(a) Economics of railways

Rail travel is becoming more popular. Fig. 1 shows the change in real GDP and the change in rail passenger journeys between 2004 and 2015.

Fig. 1 – Economic growth rate and percentage change in rail passenger journeys, 2004–2015



The railway network has a high level of fixed costs and is, therefore, often used as an example of a natural monopoly. In economic theory a natural monopoly often leads to a market which is best controlled by one firm. For many years, railway services in the UK were provided by British Rail (BR). BR was a nationalised company owned and run by the government and it had a monopoly on the provision of passenger rail services.

After 1993 there were radical changes made to the railway system in an attempt to introduce a more contestable market. The expensive infrastructure of the railway tracks and stations is now owned and maintained by Network Rail, a state owned ‘not for dividend’ company with no shareholders, which reinvests its income in the railways. Passenger services are provided by private sector train operating companies (TOCs), such as Great Western Railway, Stagecoach and Virgin. These companies are allowed to bid for the right to run a train service on a particular route. This right is in the form of a franchise that the TOC would have to pay for. The franchise lasts for a fixed time period. When franchises come up for renewal there is a bidding process, and this can result in a new TOC providing the service. To ensure that the firm operating the franchise maintains standards, a regulatory body called the Office of Rail and Road was established to oversee their activities. This body monitors compliance with health and safety legislation as well as industry targets for the rail passenger service. Targets include punctuality, availability of seating and frequency of train services. Fig. 2. shows a selection of regional rail franchises with both current and previous operators.

Fig. 2 – Selected regional rail franchises and TOCs. April 2016

Train operator (TOC)	Regional franchise	Start date of franchise	Finish date of franchise	Previous operator (TOC)
Abellio	Greater Anglia	Feb. 2012	Oct. 2018	National Express E. Anglia
Arriva UK	Wales and Borders	Dec. 2003	Oct. 2018	Wales and Borders
Stagecoach	East Midlands	Nov. 2007	Mar. 2018	Central Trains and Midland Mainline
Arriva UK	Chiltern Railways	July 1996	Dec. 2021	Network South East
Govia	West Midlands	Nov. 2007	Oct. 2017	Central Trains

The East Coast Railway Line

In 2015 an eight-year franchise for passenger train services on the East Coast mainline between London and Edinburgh was awarded to a joint venture between Stagecoach and Virgin. For the five years previous to this the provider was Directly Operated Railways, a public sector firm. Under the management of the joint venture between Stagecoach and Virgin, the franchise began with the 7.55 am service leaving Newcastle for London King's Cross.

More services have been promised for the line over the next few years, bringing a 50% increase in capacity and faster journeys to London. Stagecoach and Virgin have pledged to invest about £140m in order to deliver an improved service and a more personalised travel experience for customers. Stagecoach and Virgin are scheduled to pay £3.3bn to the government for the franchise.

The East Coast line has had several providers of passenger train services. Two private operators failed to meet their financial commitments and their franchise was removed. As a result, a small public sector firm, Directly Operated Railways, stepped in to run trains on the mainline in late 2009.

In its five years Directly Operated Railways provided over £1bn in payment to the government, as well as several million pounds in profits, to the Treasury. Financial analysis from the Office of Rail and Road shows it was one of two railway firms to make a net contribution to government, paying in more than it received in subsidy or indirect grants, along with Southwest Trains (run by Stagecoach).

The different prices charged for a train journey from Newcastle to London on the East Coast Line are shown in Fig. 3.

Fig. 3 – Price of adult single ticket from Newcastle to London King's Cross on Tuesday the 5th of April 2016

Departure time from Newcastle	Arrival time in London	Journey time	Price of adult single ticket
7.08 am	10.20 am	3 hours 12 minutes	£138.00
7.29 am	10.42 am	3 hours 13 minutes	£78.50
9.30 am	12.43 pm	3 hours 13 minutes	£51.00

Source: National Rail Enquiries

Identify and explain, using the stimulus material, one reason why the privatisation of the rail network has made it a more contestable market.

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[2]

(b)Explain, using a diagram, why the high level of fixed costs in providing the infrastructure of the railway network make it an example of a natural monopoly.

[3]

(c)

(i)What evidence is there in the stimulus material of TOCs engaging in price discrimination?

[2]

(ii)Evaluate the extent to which the consumer benefits from price discrimination by a firm with monopoly power.

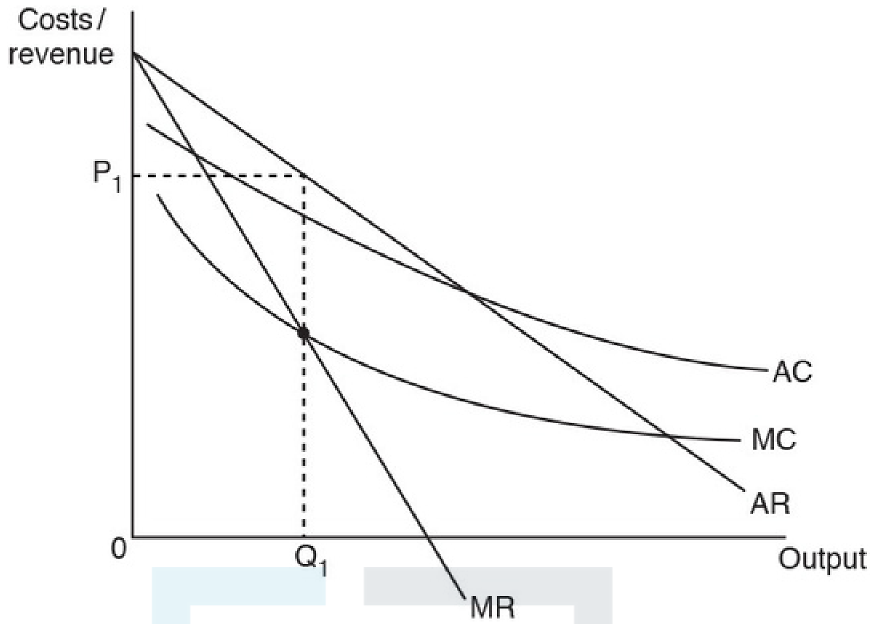
[8]

3 * Perfect competition theory is based on a set of very unrealistic assumptions.

Evaluate the usefulness of perfect competition theory in explaining the behaviour of firms in the real world.

[25]

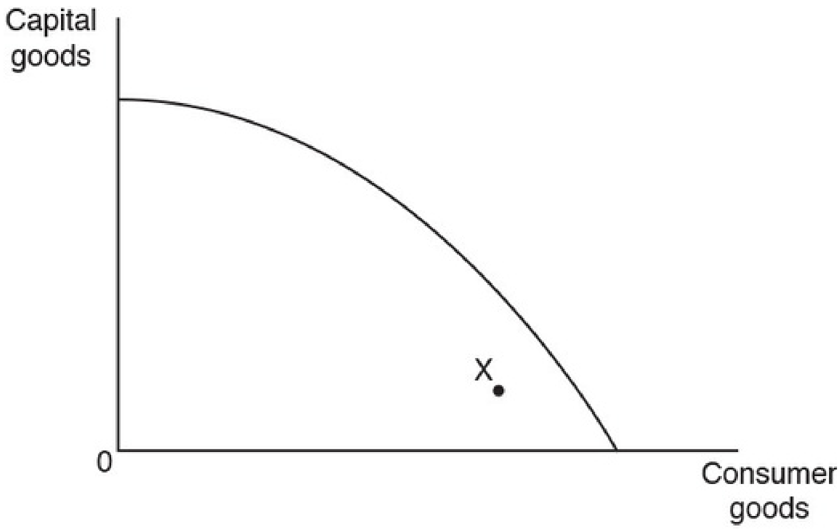
4 Which of the following is represented in the diagram below?



A Contestable market
B Monopolistic competition
C Natural monopoly
D Oligopoly

Your answer [1]

5 In the diagram below, what does point X represent?



A Allocative inefficiency
B Opportunity cost
C Productive inefficiency
D Scarcity

Your answer [1]

6 A firm could lower its average costs whilst keeping its output the same. What type of inefficiency is occurring?

A Allocative inefficiency
B Dynamic inefficiency
C Productive inefficiency
D X-inefficiency

Your answer [1]

7 Which one of the following is a defining characteristic of a contestable market?

A Collusion
B Long run supernormal profits
C Many small buyers and sellers
D No barriers to entry or exit

Your answer [1]

8

Please refer to insert H460-03 June 2018, Olympic Games, which can be downloaded separately and is also included with Question 31.

Evaluate, using an appropriate diagram, the extent to which consumer welfare is harmed as a result of the bicycle industry being an oligopoly.

[15]

9 * One of the roles of the Competition and Markets authority is to prevent the abuse of monopoly power.

Evaluate, using an appropriate diagram(s), the potential impact of monopoly power within an industry.

[25]

10 The UK labour market in the 21st century

The UK labour market has experienced both a large increase and significant changes in the past 20 years. Data published in the UK's Labour Force Survey in October 2017 shows employment continuing to rise and unemployment falling, as seen in Fig. 1.

Fig. 1 – UK labour market statistics for June to August 2017

	Number ('000)	Change on previous quarter ('000)	Change on previous year ('000)
Employed	3092	109	363
Unemployed	3	–63	–224
Economically Inactive (Aged 16–64)	1418 8812	–17	–13

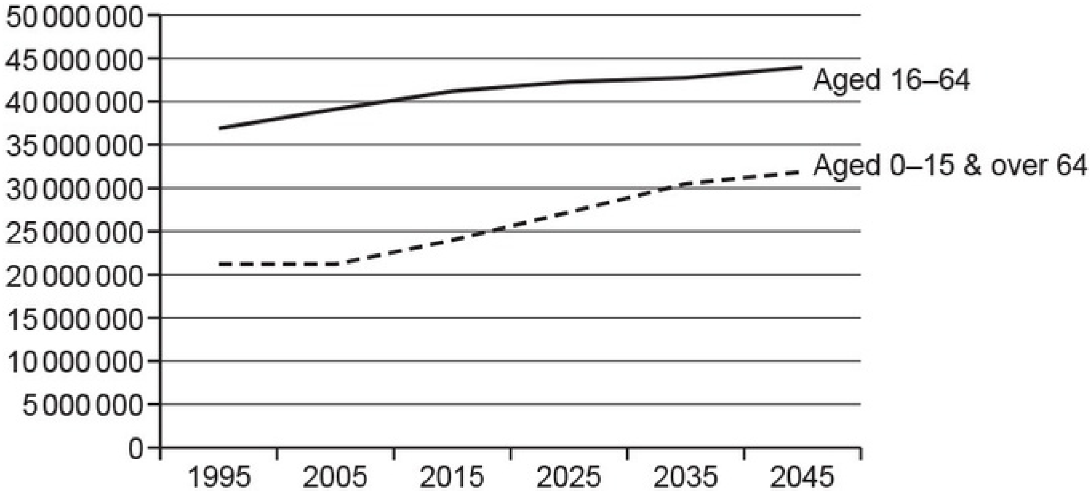
Source: ONS

The unemployment rate was at a 42-year low, although there was still some slack in the labour market. Other evidence in the October 2017 data included:

- average weekly earnings for employees increased by 2.2% in nominal terms, compared with a year earlier.
- average weekly earnings for employees decreased by 0.3% in real terms, compared with a year earlier.
- employment growth was strongest for those aged 25–34 years and 50–64 years.
- employment fell among young people aged 16–24 years.
- growth in employment over the year was greater for women (up by 246 000) than for men (up by 71 000).
- employment growth was not equal across the country, with the biggest increases in the West Midlands, London, Scotland and the South East but large falls in the South West, East Midlands and Wales.
- the proportion of workers in temporary jobs because a permanent one was not available was 27.6%.
- the proportion of people working part-time because they could not find a full-time job was 12.3%.

The UK economy has created nearly 6 million new jobs in the last 20 years which was necessary given the increase in the UK population over this period (see Fig. 2). Many of these extra jobs were primarily due to an increase in self-employment and employees in part-time work.

Fig. 2 – UK population estimates 1995–2045



Source: ONS

The 'gig economy'

Each day, just before lunch, couriers dressed in the distinctive green and black uniform of Deliveroo, the online food delivery company, arrive at the end of the street, park their bikes next to a bench and wait. Holding their smartphones, they wait for someone, somewhere in the city, to place an order with one of the nearby restaurants and cafés. When an order comes through, one of the couriers will pick it up and deliver it in exchange for a small fee. They will then return to the bench to wait for another call.

Welcome to the 'gig economy' which can be defined as, "a labour market characterised by the large number of short-term contracts or freelance work, as opposed to permanent jobs".

Plenty of people in early 21st-century Britain can identify with the experience of working for a company like Deliveroo. But even more people are employed on zero-hour contracts in a wide variety of jobs, from stacking shelves to waiting tables to caring for the elderly. According to the Office for National Statistics, around 900 000 workers rely on a job with a zero-hour contract. These people start every week not knowing how much work they will get or how much money they will earn. Informal or casual employment of this kind helps explain why the UK's unemployment rate has not grown out of control since the financial crash of 2008. This increased labour market flexibility has made it easier and less risky to employ workers, especially for monopolistically competitive firms who will expect to see a fall in their average costs.

However, opponents of the growth in casual employment, including the trade unions, claim that there are underlying problems associated with this labour model. Millions of people are "just about managing" and many are faring much worse. In the 12 months up to March 2017, the UK's largest food bank charity gave out more than one million emergency food parcels to people in desperate need. At the same time, household debt has grown to 150% of income. This debt has been fuelled by low wage growth and low interest rates which has made credit easily available. But the main issue for the estimated 8.3 million people living with unmanageable debt is needing to borrow money to survive.

It is no more than a coincidence that these concerns are happening in a year which recognises the 75th anniversary of the founding of the UK welfare state which was set up to focus on, among other issues, a problem that had grown since the early 1800s: that many workers struggled to earn regular and reliable wages throughout the entire year. However, some have come to regard the welfare state simply as a cost to be kept down rather than part of an economic and social strategy that aims to deliver security for all. Successive governments have cut benefits and altered the tax system to move the burden away from more progressive direct taxation towards more regressive indirect taxation.

Fig. 3 shows the effect on different groups of households by comparing their original income earned from employment with their final disposable income after benefits and taxation have been taken into account.

Fig. 3 – Average incomes, taxes and benefits of households by decile group, 2015/16

	Bottom	2nd	3rd	4th	5th	6th	7th	8th	9th	Top	All households
Original income	**6286**	**13988**	**17410**	**25754**	**31424**	**39293**	**47963**	**59622**	**68968**	**117960**	**42867**
+ Total cash benefits	6387	7748	8313	6319	4494	3729	2368	2283	2004	1544	4519
Gross income	**12673**	**21736**	**25723**	**32074**	**35918**	**43022**	**50331**	**61905**	**70972**	**119504**	**47386**
– Direct taxes	1528	2299	3046	4794	5940	7931	10091	13199	15534	29991	9435
Disposable income	**11145**	**19437**	**22677**	**27280**	**29978**	**35091**	**40239**	**48706**	**55438**	**89513**	**37951**

Source: ONS

Please refer to insert H460-01 June 2019, which can be downloaded separately and is included above.
A firm in a monopolistically competitive market decides to introduce more casual employment practices, such as zero-hour contracts.

Explain, using a diagram, the effect of this decision on the firm's equilibrium position in the short run.

[4]

11 * Netflix is the world's largest online video subscription service. By 2018, after only 10 years in existence, it had 137 million worldwide subscribers. 65% of online video viewers in the USA watch Netflix at least once a month.

Evaluate, using an appropriate diagram(s), the advantages and disadvantages of a perfectly contestable market.

[25]

12(a) The economics of supermarkets

Grocery retailing is often used as an example of an oligopoly market in the UK. Economists sometimes divide oligopoly markets into two types: perfect oligopoly, in which the products sold are homogenous and imperfect oligopoly, in which there are differentiated products. The behaviour of firms in a perfect oligopoly can be significantly different from those in an imperfect oligopoly.

In the UK, since the 1950s, the selling of groceries has become increasingly dominated by supermarkets. Tesco has been the market leader for a sustained period of time with over 25% of the market share. There are, however, other competitors who also maintain a significant share of the grocery market, as shown in Fig. 1.

Supermarket	Percentage share of grocery market	Percentage share of grocery market
	July 2014	July 2016
Aldi	4.8	6.2
Asda	17	15.5
Co-op	6.3	6.4
Lidl	3.6	4.5
Morrison's	11	10.7
Sainsbury's	16.6	16.3
Tesco	28.9	28.3
Waitrose	4.9	5.1
Others	6.9	7.0

Fig. 1 Percentage share of grocery market by supermarket. July 2014 and July 2016.
(As measured by % of total market sales revenue.)

The market for groceries is large with the value of sales exceeding £170bn. This has continued to grow, see Fig. 2, despite the macroeconomic problems of the UK economy and a prolonged price war between the supermarkets.

	UK grocery market by value in £bn
2014	177.4
2016	179.1

Fig. 2 UK grocery market by value £bn. 2014 and 2016.

The larger supermarkets have used price cutting in an attempt to maintain market share. While another method of differentiating their brand has been to offer better home delivery services. At present online ordering accounts for less than 10% of all grocery sales.

Sainsbury's, one of the leading supermarkets, has recently decided to integrate its business by buying out Argos.

This constitutes horizontal integration. Argos specialises in selling non-grocery products such as toys, clothing and household goods. These non-grocery products are part of Sainsbury's wider product portfolio. Argos offers delivery by 10pm for any of its products ordered by 6pm on the same day. One of the potential benefits of this buy out of Argos by Sainsbury's is the delivery network that Argos has established. Sainsbury's hope to increase their ability to deliver food using online ordering. However, the delivery of items such as clothes and household goods creates fewer problems than delivering fresh food products. Food items require different temperatures and items are often fragile or bulky making the process more difficult and costly.

The major problem with online grocery sales is the lack of profitability. There is a very high level of fixed costs involved. It has been estimated that, on average, a delivery costs the supermarket £20 whereas consumers are charged less than £5. Ocado, an online groceries only provider, operated for 15 years before making a profit.

In recent years there have been some successful new entrants to the grocery retailing market such as Aldi, Lidl and Netto. Most recently, Amazon, a market leader in online sales, has also entered the grocery retailing market through a new service called Pantry. The entry of new firms into a market can influence the profits earned by the incumbent firms. Fig. 3 shows how the profits earned by Tesco and Sainsbury's have changed in recent years.

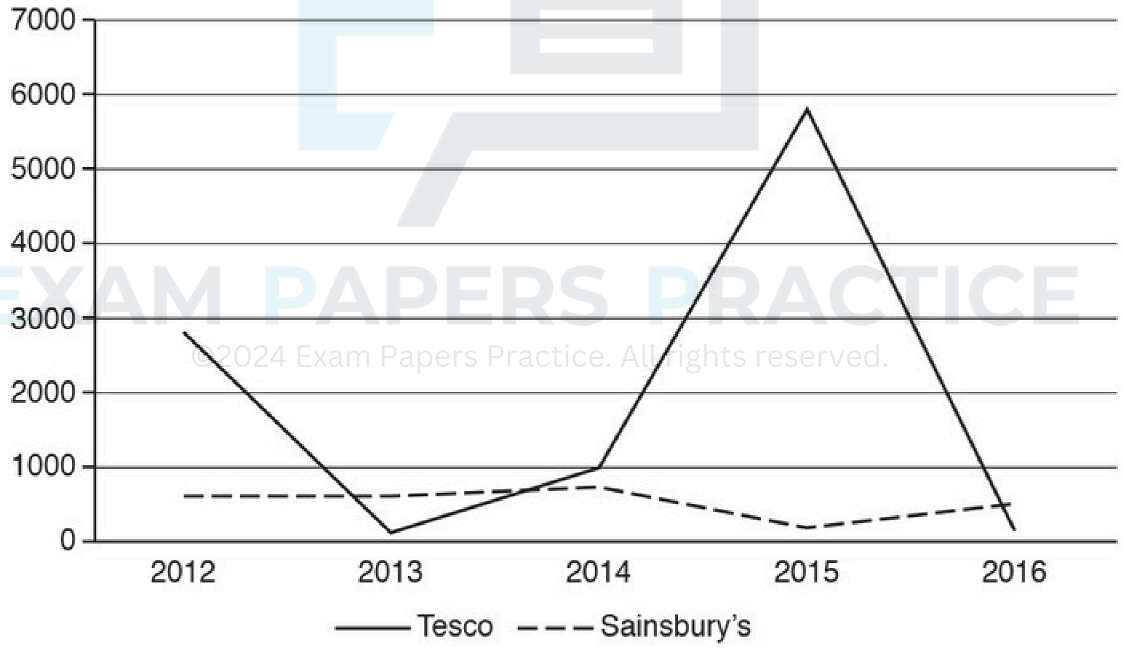


Fig. 3 Profits (£m) earned by Tesco and Sainsbury

Please refer to Insert H460-01 Practice 2, which can be downloaded separately and is also included above.

Compare the 3 firm concentration ratio for supermarkets in the grocery market in July 2014 and July 2016.

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[2]

(b)Explain, using a diagram, the significance of the kinked demand curve model on the way in which firms behave in oligopoly markets.

[4]

(c) * Using evidence from the stimulus material, evaluate the extent to which the grocery market in the UK can be described as an oligopoly.

[12]

13 * A number of industries in the UK are said to operate under conditions of monopolistic competition.

Monopolistic competition, with its focus on a large number of producers, is a realistic model of market competition.

Evaluate, using an appropriate diagram(s), the advantages to the consumer of monopolistic competition.

[25]

14(a)

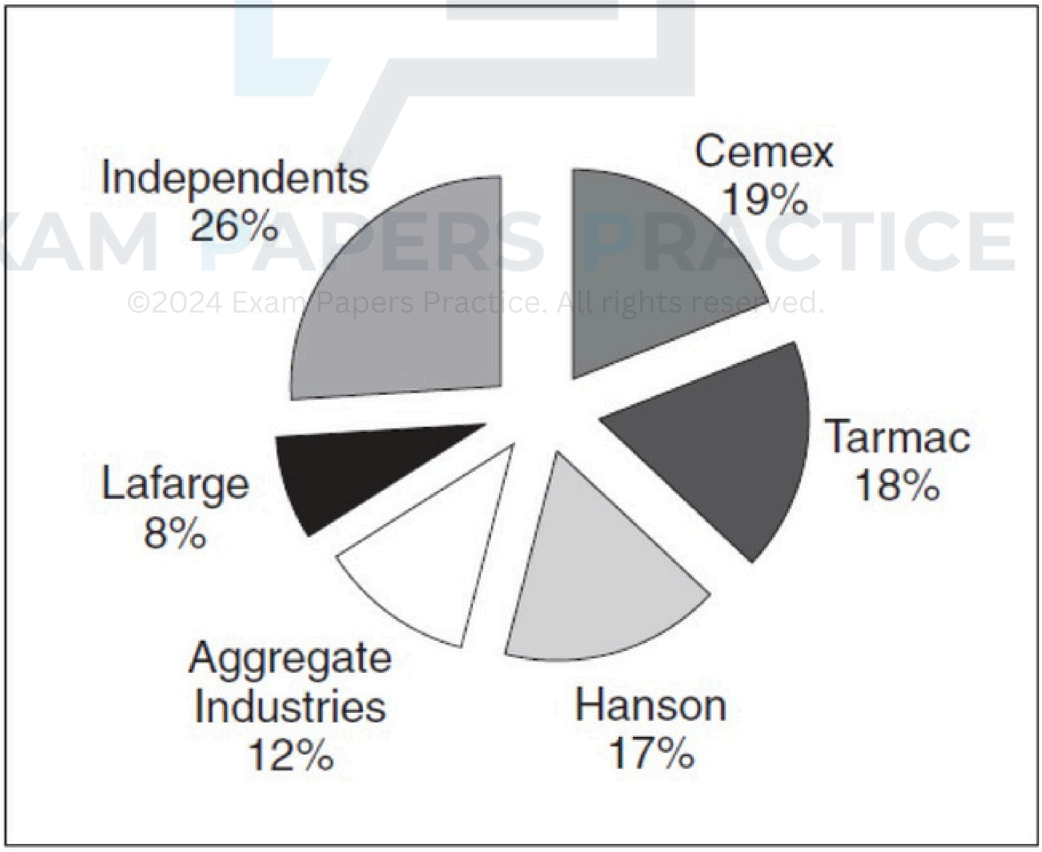
Developments in the UK cement and ready–mix concrete markets

Cement is mainly used to make concrete and accounts for approximately 45% of its production cost. The most common type of concrete is ready–mix concrete. This is a mixture of cement, granular raw materials and water. In 2009 the sales of cement in the UK totalled £0.9bn and the sales of ready–mix oncrete came to £1bn. In the same year the output of cement was 7.6m tonnes and the output of ready–mix concrete was 14m tonnes.

Over the last 20 years, a high number of mergers have reduced the number of firms in both markets and combined firms producing both cement and ready–mix concrete. In 2009 the three largest firms in the cement market, Lafarge, Cemex and Hanson, produced 6m tonnes of concrete between them.

Large firms also play a key role in the market for ready–mix concrete. Fig. 1 shows the market shares of the five largest ready–mix firms in 2009.

Fig. 1 – UK shares of ready–mix concrete volumes, 2009



The large firms, all of which are multinationals, concentrate on producing building materials. Each owns quarries which extract granular raw materials. Their factories involved with the production of concrete sell some of their output internally to other parts of their business. Some is also sold to other firms, including small firms which just produce ready–mix concrete.

There is little difference in the quality of both the cement and ready–mix concrete produced by the different firms. In the markets for both cement and concrete the large firms have considerable information about the pricing and output strategies of their rivals. This information comes from, for example, buying and selling to each other and details provided in planning proposals.

In both the markets for cement and concrete, there are a number of significant barriers to entry and exit which give incumbent firms protection from the potential entry of new firms. There are high set up costs with, for instance, a cement factory costing as much as £100m to build. In addition, there are thought to be significant economies of scale available to the larger operators.

In 2010 and 2011 the Office of Fair Trading (OFT) undertook a study of the cement and ready–mix concrete markets. There was concern about the lack of competition and possible co–ordination of market strategies undertaken by the large firms. Small ready–mix concrete firms told the OFT that they were being charged high prices for cement by the large firms, while having to compete against the low prices the large firms were charging for their own ready–mix concrete. They suggested that the large firms were prepared to accept lower profit margins on ready–mix concrete knowing that they could still make a high level of supernormal profit producing cement. These smaller firms claimed that such a strategy, which became particularly evident in 2009, was designed to drive them out of business. Figs 2 and 3 show how the average revenue and average cost changed in the markets for cement and concrete between 2006 and 2010.

Fig. 2 – Cement – average revenue and average cost per tonne

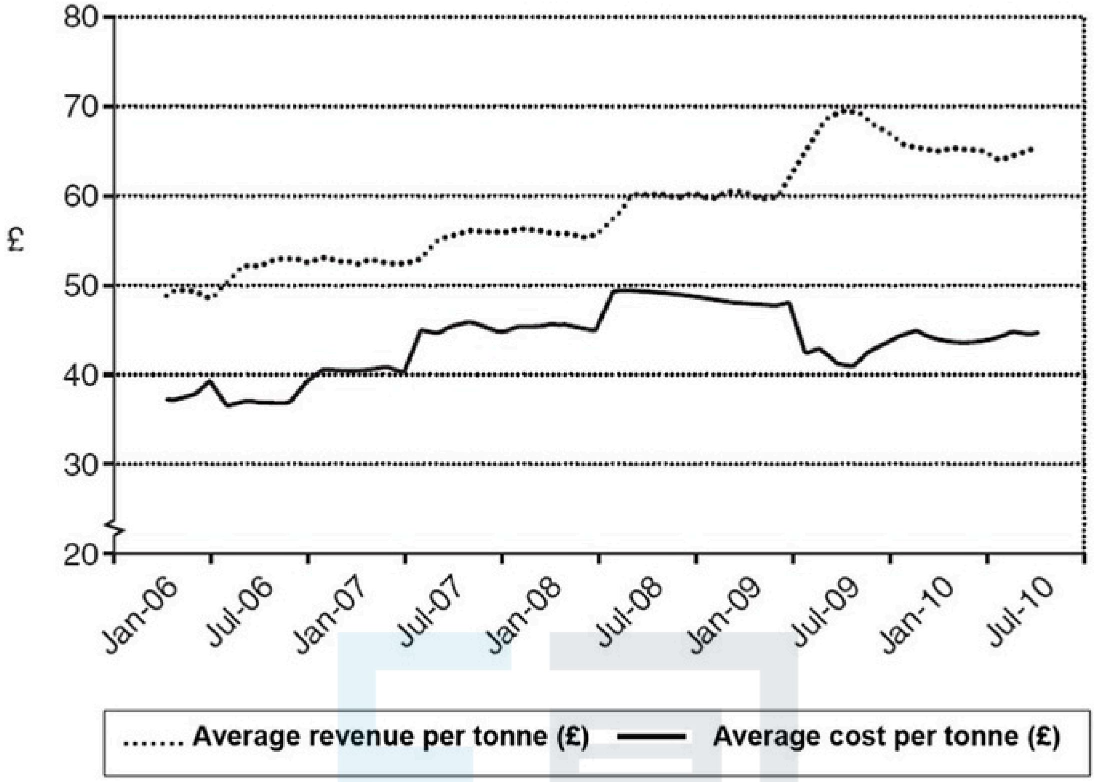
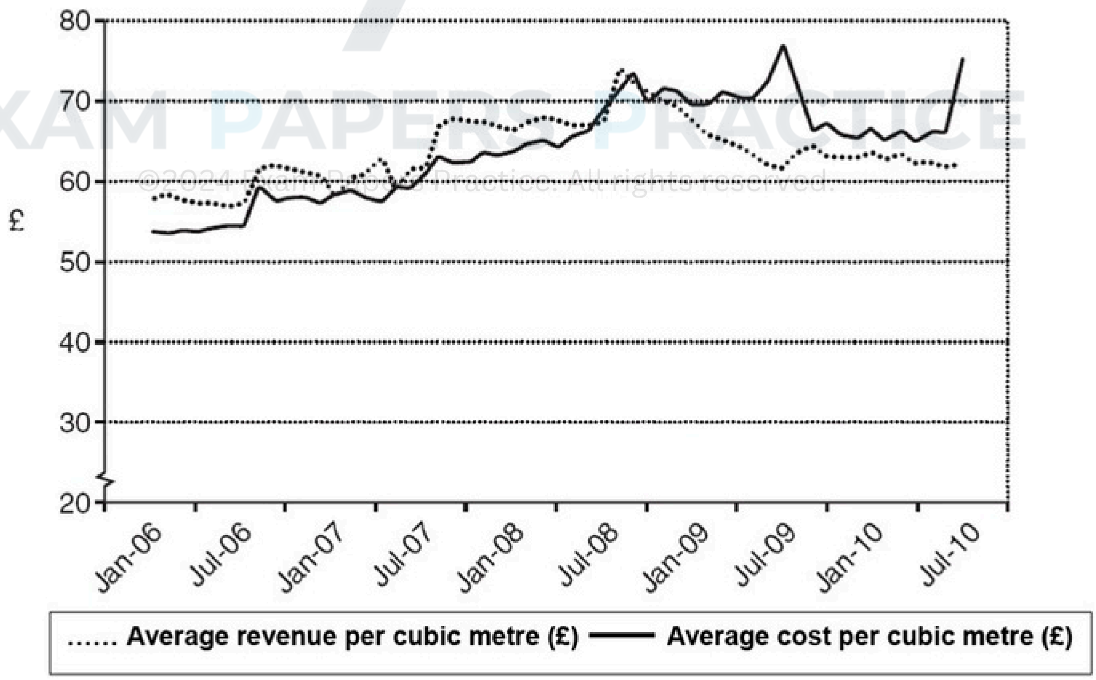


Fig. 3 – Ready–mix concrete – average revenue and average cost per cubic metre



In 2013 a merger took place between Lafarge and Tarmac but to gain permission from the regulatory authorities for this merger, the firms had to agree to sell off some of their quarries and factories to a new entrant, Hope Construction Materials. In early 2014 the Competition Commission ordered the newly merged firm, Lafarge Tarmac, to sell off more factories to create another major entrant.

It is likely that future years will see further changes in the markets for cement and concrete – nothing is set in stone.

Please refer to insert H460-01 Specimen which can be downloaded separately and is included above. Explain, using relevant calculations, whether the cement market was more concentrated or less concentrated than the ready–mix concrete market in 2009.

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[4]

(b)* Using evidence from the stimulus material, evaluate the extent to which collusion occurred in the UK cement market.

[12]

15 * In 2014, Telefonica, a Spanish telecommunications firm with a very large share of the Spanish market, decided to concentrate largely on increasing its sales revenue.

Evaluate, using an appropriate diagram(s), the extent to which a monopolist changing its objective from profit maximisation to sales revenue maximisation would benefit consumers.

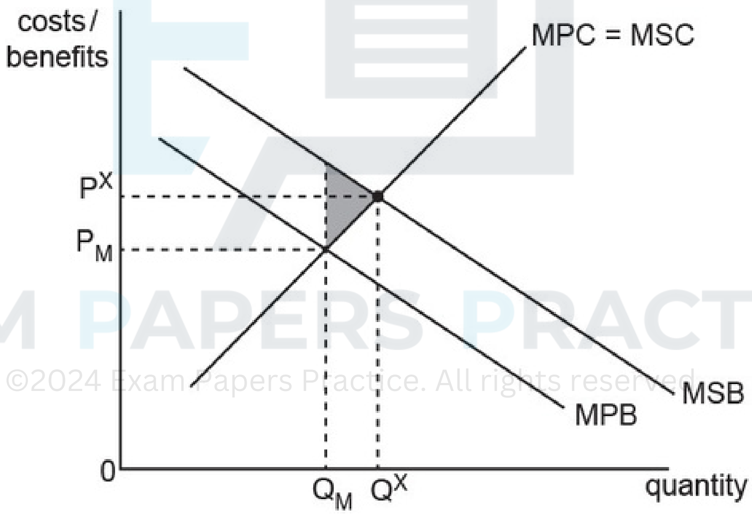
[25]

16 Which of the following identifies the concept of static efficiency?

Option	Allocatively Efficient	Dynamically Efficient	Productively Efficient	X-Efficient
A	✓	x	✓	x
B	✓	x	x	✓
C	✓	x	✓	✓
D	x	✓	✓	✓

Your answer [1]

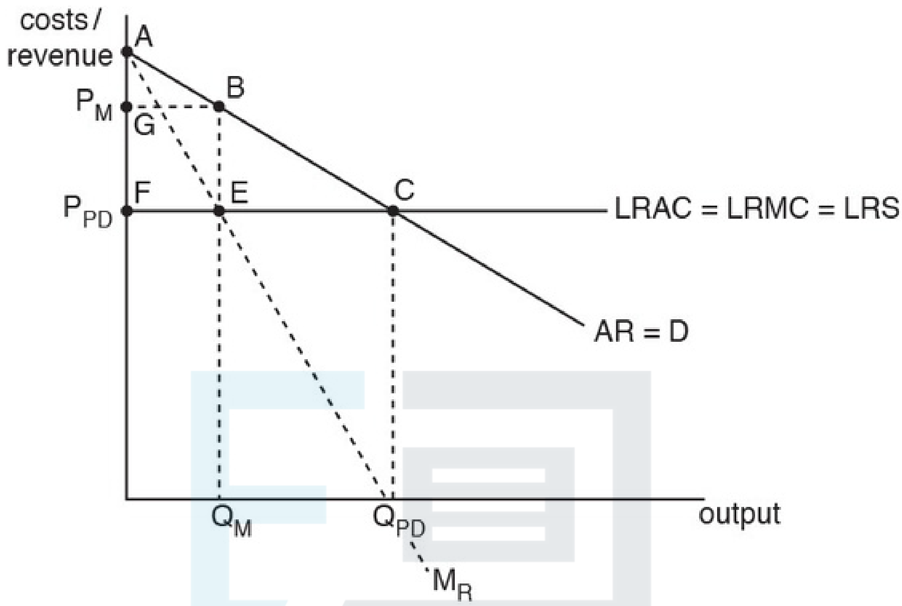
17 Which of the following options explains what is represented by the shaded area below?



A Allocative inefficiency
B Dynamic inefficiency
C Productive inefficiency
D X-inefficiency

Your answer [1]

18 In the diagram below, QM represents the output of a monopolist which is not price discriminating. If the firm was able to engage in perfect (first degree) price discrimination it would produce an output of QPD. Which area represents the loss in consumer surplus that arises as a result of this change?



A ABG

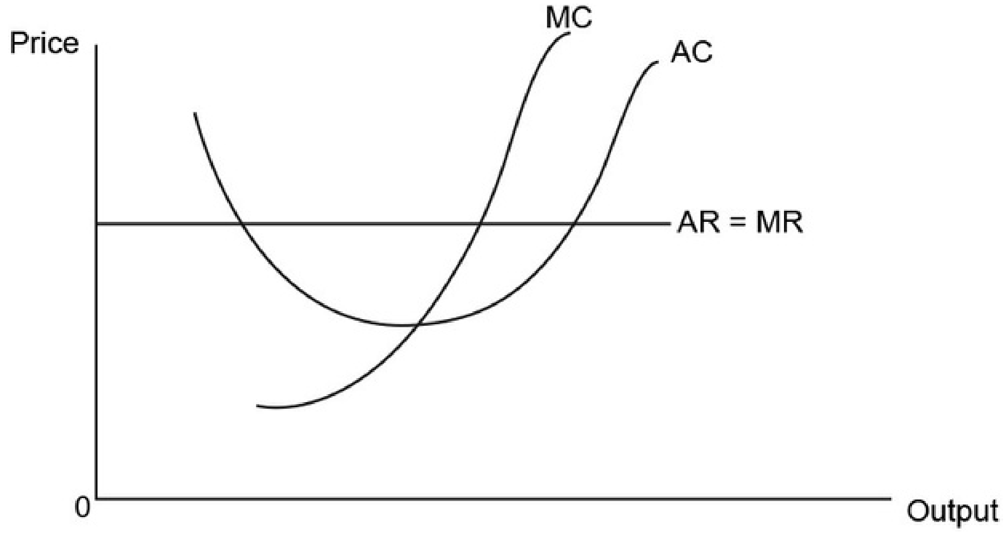
B ACF

C BCE

D BEFG

Your answer [1]

19



Which of the following options is represented by the diagram?

A Monopolistic competition in the long run
B Monopolistic competition in the short run
C Perfect competition in the long run
D Perfect competition in the short run

Your answer [1]

20 How do the characteristics of an oligopoly market differ from a monopoly market?

A Firms are short run profit maximisers
B Supernormal profits are possible in the long run
C There are a few dominant firms
D There are high barriers to entry and exit

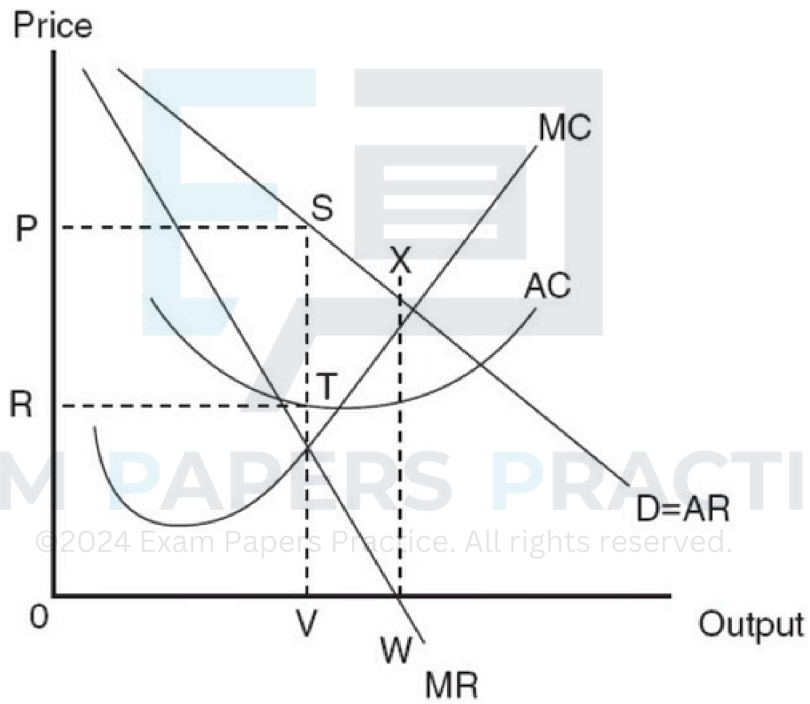
Your answer [1]

21 When is a firm dynamically efficient?

A Investment in R&D is enabling long run average costs to be reduced in the future.
B Price is equalling marginal cost.
C Production is taking place at the minimum efficient scale.
D Production is taking place on the lowest attainable average cost curve.

Your answer [1]

22 The diagram below represents a firm operating in monopolistic competition.



Which of the following statements is correct?

A The firm achieves productive efficiency
B The firm earns total revenue represented by the area RPST
C The firm is operating on the elastic section of its demand curve
D The firm's supernormal profit is represented by the area 0RTV

Your answer [1]

23 In which of the following cases are firms most likely to be productively and allocatively efficient?

A Monopolistically competitive market in the long run with firms making normal profit
B Natural monopoly
C Perfectly competitive market in the short run with firms making supernormal profit
D Perfectly contestable market

Your answer [1]

24 Two firms, Y and Z, are competing in an oligopoly market. They are deciding whether to price their good at £10 or £5. Game theory can be used to analyse their behaviour. If collusion breaks down, which of the options below represents the profits firms will earn in the long run?

		Firm Z	
		£10	£5
Firm Y	£10	Option A £150m, £150m	Option B £30m, £250m
	£5	Option C £250m, £30m	Option D £80m, £80m

Your answer [1]

25 What is most likely to lead to allocative efficiency being achieved in an industry that is a natural monopoly?

A Collusion
B Competition policy
C Information provision
D Regulation

Your answer [1]

26 What efficiencies occur in the model of perfect competition in the long run?

A Allocative and dynamic efficiency
B Allocative, dynamic and productive efficiency
C Allocative, productive and X efficiency
D Allocative, dynamic, productive and X efficiency

Your answer [1]

27 Using the table below, what has happened to the three firm market concentration ratio in the global mobile telephone market between 2010 and 2019?

Firm	2010 Market Share (%)	2019 Market Share (%)
Apple	33.2	24.8
Huawei	0	10.6
Nokia	37.6	0.4
Others	18.8	32.9
Samsung	2.4	31.3

A Decreased by 0.6%
B Decreased by 6.1%
C Decreased by 6.5%
D Decreased by 16.7%

Your answer [1]

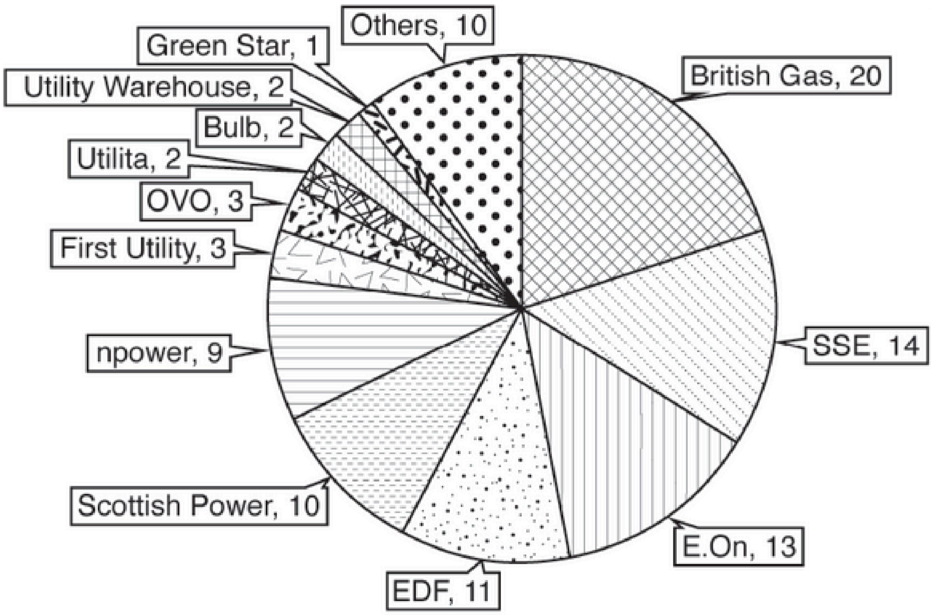
28 A bright future in Uganda

Headmaster Samuel Mukisa is one of millions of people in Uganda who are plugging into the global solar-power boom. As his students study, a single rooftop solar panel generates power. It is a solar storage system which costs Mukisa a daily rate of 15 cents. The average Ugandan earns as little as $1.50 (£1.14) a day. He pays his bill each fortnight and will do so for another year. When the payment plan is completed, Mukisa intends to upgrade to a second solar panel, so another class of students can shine.

For the 600m Africans living without access to power, fuel poverty is a socio-economic barrier. The World Bank estimates that if sub-Saharan Africa's economies had dependable electricity, GDP growth across the region could be up to 2% per annum higher than current rates.

Uganda, with a population of 42m, has one of the youngest populations on average in the world, with 7 out of 10 people under the age of 25. Incomes in the country have increased by 26% in the last two years. The service sector has grown from 32% of the economy in 1990 to 55% in 2018, with a 15% expansion in the last two years. Today, it is estimated that 44% of Ugandans use the Internet, 70% have a mobile phone and 6% use social media. The growing demand for energy in Uganda is clear. Nevertheless, away from large cities, many individuals still operate in a barter economy. Rural agriculture based areas are still very common. However, their contribution to GDP is now only 24% of the total, down from 56% in 1990; a further reflection of the developing nature of the economy.

The energy supply market in Uganda, as in many African countries, is highly localised, with only 15% of its citizens living in homes connected to the country's energy network. The rest have to rely on arranging their own supply, increasingly using solar power. This is very different to the UK, where most of the population are customers of one of the large national energy suppliers (see Fig. 1).



Source: OFGEM

Fig. 1 - UK energy suppliers' market share, 2018 (%)

Uganda is at the centre of the solar energy revolution in Africa. For development organisations, including the United Nations, and socially minded investors, the opportunities are exciting. Street lights, topped with solar panels, charge up through the bright daylight hours and illuminate the streets each night. For a growing number of energy companies, the huge number of people without reliable power represents the opportunity to access a growing market.

Local start-up firms have appeared quickly, many with the backing of international companies. In 2018, one multinational energy firm, Shell, invested in three start-ups, SolarNow, SteamaCo and Husk Power Systems. These three have raised a total of $30m from Shell and other investors. Shell's charity foundation also backs a UK-based firm, BBOXX. BBOXX has expanded rapidly, providing clean energy solutions and pay-as-you-go solar power. It aims to have provided home solar power systems to 20m people in Africa by 2020.

If Shell hopes to gain a control in the market it will need to compete against a former rival. The French firm, Engie, has taken the most decisive step into Africa's home solar market, with the acquisition of Ugandan-based Fenix International. Fenix supplies power to around 250 000 homes in Uganda, through the solar kits it sells in partnership with a mobile network provider.

The popularity of the Fenix home energy system means that the company is growing. It plans to expand into other countries, to reach millions more people. "We intend to be in multiple markets across eastern, southern and western Africa," said Chief Executive, Lyndsay Handler. "It's still amazing to me that so many people in Uganda do not have access to power. We provide a decentralised, affordable home system that can completely transform lives. An opportunity to rethink the way that we can achieve universal access to energy," she said.

"Beyond wanting to have an impact, we have always wanted to prove to the world that you can have a positive impact and deliver commercial returns. There's an underlying drive to show that this is not just charity. You can build commercial returns in Africa. We want to be part of a story that encourages others to see Africa as a commercial market. There are entrepreneurial people, with ideas and opportunities. They need investment, not just handouts," she said.

One Fenix customer for the past three years is Ritah Namatovu. She is now able to work at night to support her four grandchildren, thanks to her solar energy kit. Namatovu's only concern is that the solar panel might one day be stolen from her roof. "The solar panel is an asset I now own. There are no blackouts. These panels look so small, but they are powerful," she said.

Please refer to insert H460-01 October 2020, which can be downloaded separately and is included above.

Refer to Fig. 1.

Calculate the five-firm concentration ratio in the UK energy supplier market.

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29 * In recent years, most towns and cities have seen a rapid growth in the number of coffee shops and sandwich bars, the majority of which are small and locally-owned.

Evaluate, using an appropriate diagram(s), whether such small firms, operating in a monopolistically competitive market, can be economically efficient.

[25]

END OF QUESTION PAPER